The optimization process for a single stock order and trade portfolio is shown below. The single stock process is further discussed in Chapter 8, Algorithmic Decision Making Framework and the portfolio optimization process is further discussed in Chapter 9, Portfolio Algorithms.

Single Stock Trade Cost Objective Function

$$Min(b_1 I^* \alpha^{a4} + (1-b_1)I^*) + \lambda \cdot \left(\sigma \cdot \sqrt{\frac{1}{250} \cdot \frac{1}{3} \cdot \frac{S}{ADV} \cdot \frac{1}{\alpha}} \cdot 10^4 bp \right) \tag{7.57}$$

Portfolio Trade Cost Objective Function

$$Min \sum_{i=1}^{m} \sum_{j=1}^{n} x_{ij} \cdot \left(b_1 \cdot I_i^* \cdot \left(\frac{x_{ij}}{X_i} \right) \cdot \frac{x_{ij}}{v_{ij}} + \left(\frac{x_{ij}}{X_i} \right) \cdot (1-b_1) \cdot I_i^* \right) + \lambda \cdot \sqrt{\sigma^2 \cdot \sum_{j=1}^{n} r_k' \, Cr_k} \tag{7.58}$$

Both of these optimizations will also contain user specified constraints.

Author's Note:

It is important to mention that the parameter λ is used to specify the investor's level of risk aversion. This represents how much market impact cost the investor is willing to incur to reduce timing risk by an additional unit. In this formulation lambda can take on any value greater than zero. That is, $\lambda \geq 0$.

In a cost-risk optimization the value of lambda is directly related to the resulting optimal trading strategy plotted on the efficient trading frontier. The tangent on the ETF at this point will be equal to the negative value of specified lambda. If $\lambda = 1$ then the tangent of the ETF at this point will have a slope equal to $m = -1$.

In general, setting lambda high will result in an aggressive strategy with higher market impact but lower timing risk. Setting lambda low will result in a passive strategy with lower market impact but higher timing risk. Unfortunately, there is no universal convention for the meaning of lambda in the optimization process.

Some brokers will optimize the trade-off between cost and variance rather than cost and standard deviation (as we show above). In optimization, the meaning of lambda will be much different using variance rather than standard deviation (the square root of lambda). Additionally, the